

Project Initialization and Planning Phase

Date	07 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed project aims to automate the credit card approval process using Machine Learning, improving prediction accuracy and reducing manual effort. The system analyzes applicant information to predict whether a credit card application is likely to be approved or rejected. It enables faster decision-making, improves operational efficiency, and enhances customer experience through an intelligent web-based application.

Project Overview	
Objective	The primary objective is to develop a Machine Learning-based Credit Card Approval Prediction System that provides fast, accurate, and reliable approval predictions using applicant information.
Scope	The project focuses on analyzing applicant demographic, employment, financial, and family information to predict credit card approval. It includes data preprocessing, model training, evaluation, and deployment using a Flask web application.
Problem Statement	
Description	Manual credit card approval is time-consuming, inconsistent, and prone to human errors. Financial institutions require an intelligent system that can automate the approval process while maintaining high prediction accuracy.
Impact	Automating credit card approval improves operational efficiency, reduces processing time, minimizes human errors, and enables faster decision-making, resulting in better customer satisfaction.

Proposed Solution	
Approach	Develop a Machine Learning model using Logistic Regression to analyze applicant details and predict whether a credit card application is approved or rejected. Deploy the trained model through a Flask web application for real-time predictions.
Key Features	- Machine Learning-based credit card approval prediction. - Real-time prediction through a Flask web application. - High prediction accuracy (98.31%). - Simple and user-friendly interface. - Fast and reliable decision support.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU	Intel Core i5 (or equivalent)
Memory	RAM	8 GB
Storage	Disk space	20 GB Available Storage
Software		
Frameworks	Backend Framework	Flask
Libraries	Python Libraries	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, XGBoost

Development Environment	IDE	Jupyter Notebook, Visual Studio Code
Data		
Data	Source, size, format	Kaggle Credit Card Approval Dataset (application_record.csv & credit_record.csv), CSV Format